TaskQueue Technical Reference

1. Delivery guarantees

TaskQueue provides at-least-once delivery. A task is handed to a worker, and the worker must acknowledge it before the visibility timeout expires. If a worker crashes before acknowledging, the task becomes visible again and another worker receives it, so a task can be delivered more than once. Handlers must therefore be idempotent.

2. Retry and backoff

Retries use exponential backoff. The delay before attempt n is base * 2^n seconds, capped at 300 seconds. Full jitter then picks a random delay between zero and that value, which spreads retries out and prevents many failed tasks from hitting a dependency at the same moment (the thundering herd problem). After the maximum number of attempts a task stays in the failed set.

3. Persistence and compaction

The SQLite backend runs in write-ahead logging mode. Completed tasks are retained for seven days and are then deleted by compaction. Deleting rows does not shrink the database file; a VACUUM rewrites the file and returns the unused space to the operating system.